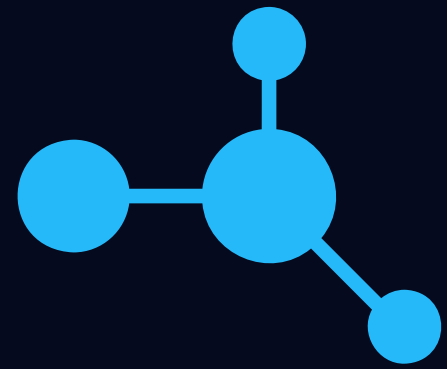




ARTCONNECT APPLICATION

Christine Khazaka

DATABASE PROJECT

PROJECT OBJECTIVE

1

Transform ArtConnect from an in-memory Java app into a database-connected application.

3

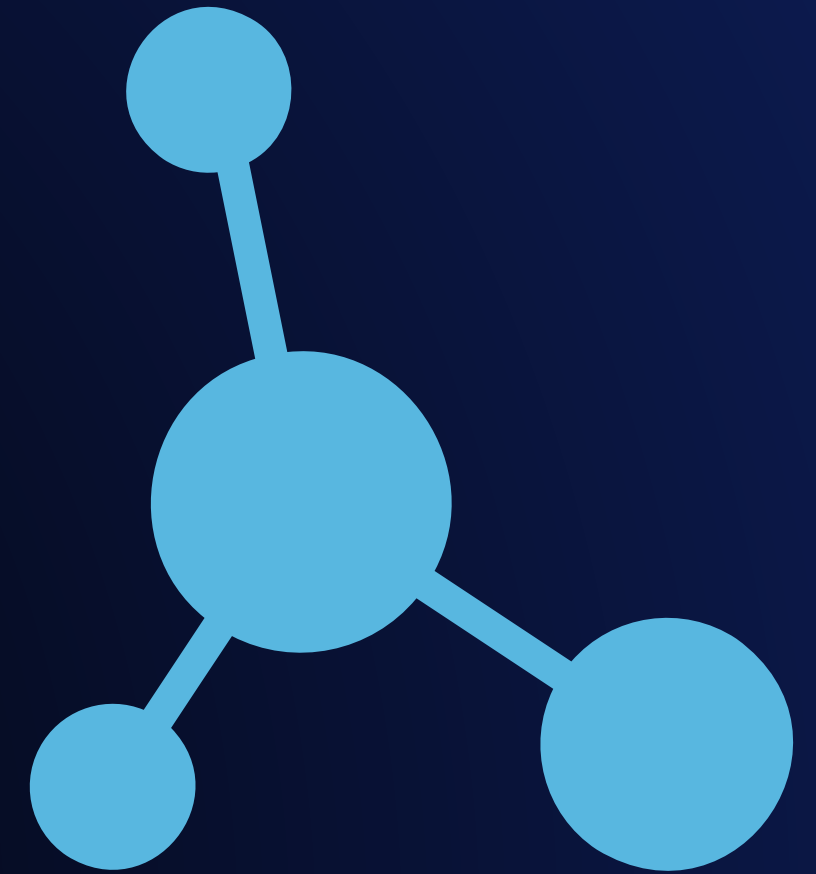
Implement SQL features such as views, indexes, triggers, procedures, and transactions.

2

Design a relational database for users, artists, artworks, galleries, exhibitions, workshops, and bookings.

4

Connect the Java application to MySQL using JDBC, DAO, and service layers.



MAIN FEATURES

- Browse featured exhibitions and workshops.
- Manage artists, artworks, galleries, exhibitions, workshops, and community members.
- Allow members to book workshops.
- Allow members to follow artists.
- Link artworks to exhibitions.
- You can add one screenshot of the app here.



USER ROLES

ROLE

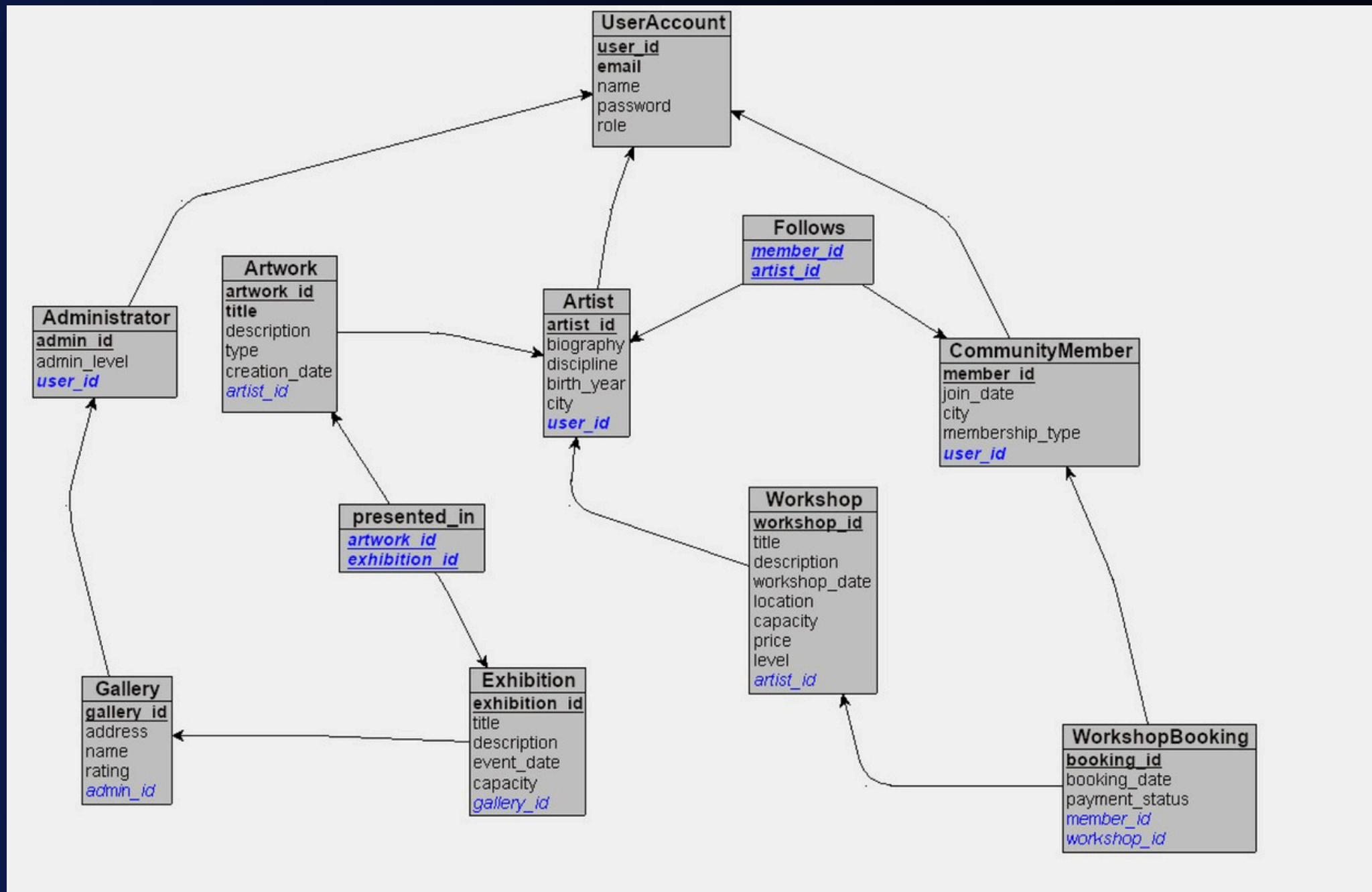
- Community Member
- Artist
- Administrator

MAIN ACTIONS

- Browse content, follow artists, book workshops
- Create artworks, organize workshops
- Manage galleries and platform data



MAIN DATABASE TABLES



- UserAccount
- CommunityMember
- Artist
- Administrator
- Artwork
- Gallery
- Exhibition
- Workshop
- WorkshopBooking
- Follows
- presented_in

VIEWS, INDEXES, TRIGGERS, PROCEDURES

Views:

- vw_users_sanitized: hides passwords.
- vw_exhibition_summary: simplifies exhibition display.
- vw_member_workshop_bookings: shows member booking history.

Triggers

- prevent invalid workshop capacity.
- prevent booking full workshops.
- automatically reduce capacity after booking.
- prevent duplicate follows.

Indexes:

- Improve search by city, discipline, date, and foreign keys.

Procedures:

- create workshop.
- book workshop.
- follow artist.
- cancel booking.

JAVA APPLICATION ARCHITECTURE

UI Controllers



Service Layer



DAO Interfaces



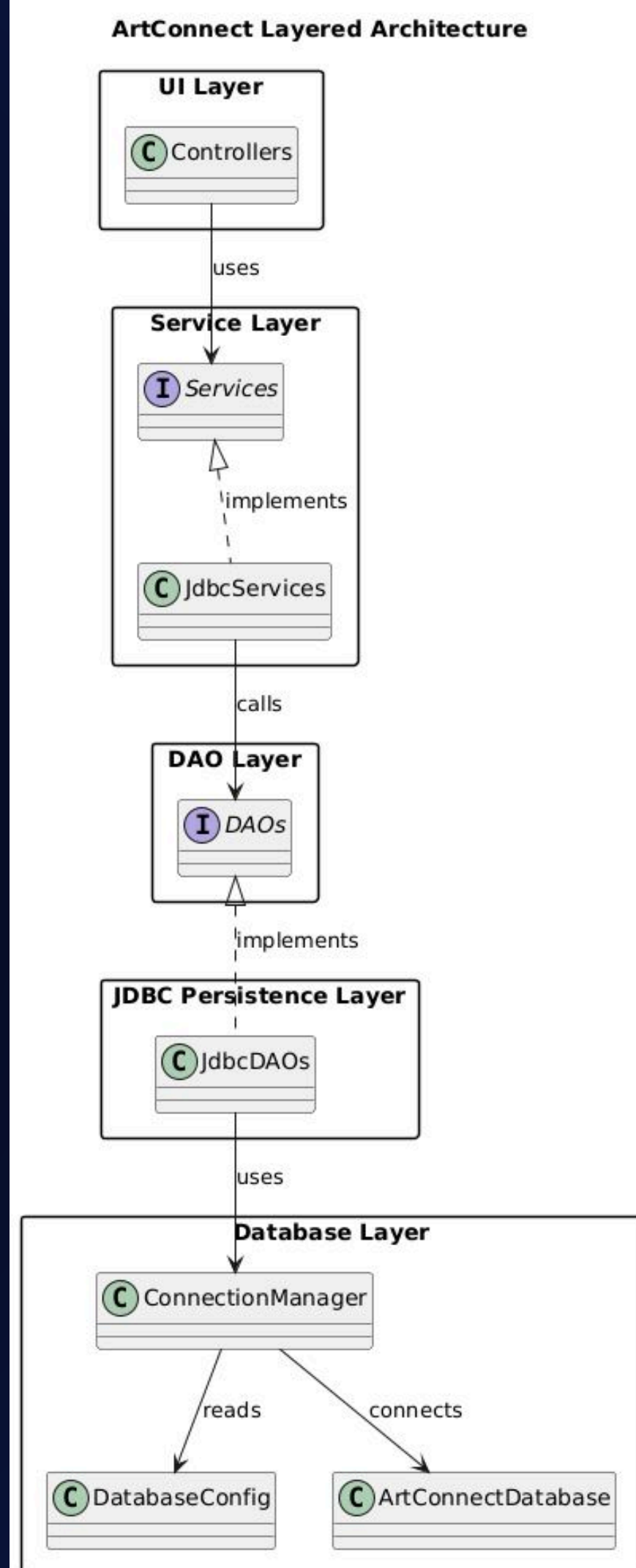
JDBC DAO Implementations



ConnectionManager / DatabaseConfig



MySQL ArtConnect Database



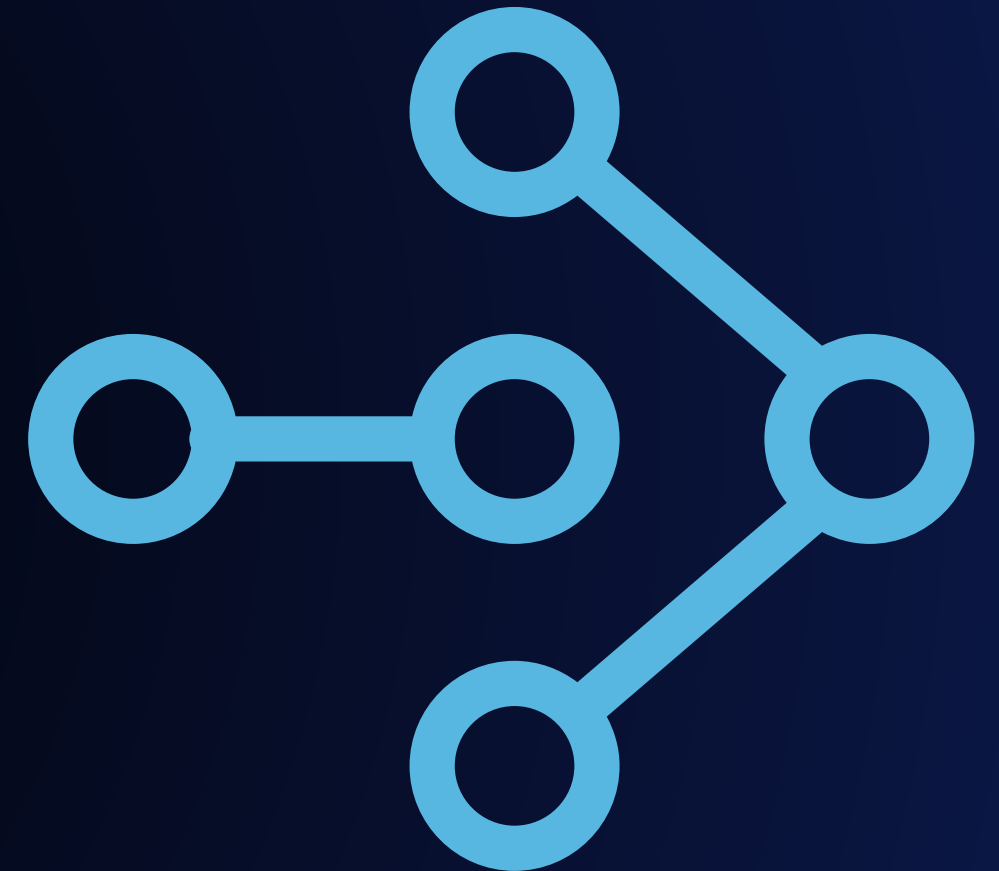
TESTING

01. TESTING OF THE CRUD

02. TESTING OF THE CONSTRAINTS

03. TESTING OF THE TRIGGERS

04. TESTING OF THE LINK BETWEEN THE APP AND THE DATABASE



CONCLUSION

CHALLENGES AND DIFFICULTIES :

- replacing in-memory services with JDBC persistence;
- keeping the UI connected to the new service layer;
- handling relationships between tables;
- preventing invalid database operations.

- ArtConnect is now connected to a MySQL database.
- The database supports persistent CRUD operations.
- The project uses a clean layered architecture.
- SQL advanced features improve reliability, security, and performance.

